



‘येथे बहुतांचे हित ।’

Marathwada Mitramandal's COLLEGE OF ENGINEERING

Karvenagar, Pune

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Unit 2: Testing and Data Modeling (06 Hours)

Random Numbers and Simulation: Sampling of continuous distributions, Monte Carlo methods
Hypothesis Testing: Type I and II errors, rejection regions; Z-test, T-test, F-test, Chi-Square test, Bayesian test

Stochastic Processes and Data Modeling: Markov process, Hidden Markov Models, Poisson Process, Gaussian Processes, Auto-Regressive and Moving average processes, Bayesian Network, Regression, Queuing systems

Session No. 1

Random Numbers and Simulation: Sampling of continuous distributions

Random Numbers

Random numbers are values generated in such a way that each value is unpredictable and independent of others. They are essential for simulations because they allow for the creation of scenarios that mimic the variability and unpredictability of real-life situations.

Types of Random Number Generators:

1. **True Random Number Generators (TRNGs):** These rely on physical phenomena (like electronic noise) to generate randomness.
2. **Pseudo-Random Number Generators (PRNGs):** These use mathematical algorithms to produce sequences of numbers that appear random. PRNGs are deterministic but can generate sufficiently random-like sequences for practical purposes.

Key Properties of PRNGs:

- **Periodicity:** The sequence eventually repeats after a certain length, known as the period.
- **Seed:** An initial value used to start the sequence. Different seeds produce different sequences.
- **Uniformity:** The numbers should be uniformly distributed over the desired range.

Simulation

Simulation is the process of creating a model of a real system and conducting experiments on this model to understand the behavior of the system or evaluate strategies for its operation.

Sampling from Continuous Distributions

Sampling from continuous distributions involves generating random numbers that follow a specific probability distribution, such as the normal, exponential, or uniform distribution. Here are some common techniques:

1. Inverse Transform Sampling:

- Applicable when the cumulative distribution function (CDF) $F(x)$ of the target distribution is known.
 1. Generate a uniform random number u between 0 and 1.
 2. Find x such that $F(x) = u$. This x is a sample from the desired distribution.
- Example: For an exponential distribution with rate λ , the CDF is $F(x) = 1 - e^{-\lambda x}$. Solving $u = 1 - e^{-\lambda x}$ gives $x = -\frac{\ln(1-u)}{\lambda}$.

Rejection Sampling:

- Useful when the target distribution is complex or when $F(x)$ is not easily invertible.
- Steps:
 1. Generate a proposal sample y from a simpler distribution $g(y)$.
 2. Generate another uniform random number u .
 3. Accept y as a sample if $u \leq \frac{f(y)}{Mg(y)}$, where $f(y)$ is the target distribution and M is a constant such that $f(y) \leq Mg(y)$ for all y .
- Otherwise, repeat the process.

Box-Muller Transform:

- Specifically used for generating normally distributed (Gaussian) random numbers.
- Steps:
 1. Generate two independent uniform random numbers u_1 and u_2 between 0 and 1.
 2. Compute $z_0 = \sqrt{-2 \ln(u_1)} \cos(2\pi u_2)$ and $z_1 = \sqrt{-2 \ln(u_1)} \sin(2\pi u_2)$.
 3. z_0 and z_1 are independent standard normal random variables.

Applications of Random Numbers and Simulation

- **Monte Carlo Methods:** Utilize random sampling to estimate numerical results. Commonly used in finance for option pricing, risk assessment, etc.
- **Stochastic Processes:** Simulate random processes that evolve over time, such as stock prices or queues in a network.
- **Optimization:** Techniques like simulated annealing use random sampling to find optimal solutions.

Monte Carlo methods

Monte Carlo methods are used to model phenomena with significant uncertainty in inputs and systems with a large number of interacting components. They are named after the Monte Carlo Casino in Monaco, reflecting the element of chance at the core of these methods.

Key Concepts

1. **Random Sampling:**
 - The core of Monte Carlo methods is the generation of random samples.
 - These samples are used to explore the possible outcomes of a system or process.
2. **Stochastic Models:**
 - Models used in Monte Carlo simulations often include random variables to capture uncertainty and variability.
 - The behavior of these models is studied through the generated random samples.
3. **Repetition:**
 - A large number of simulations are run to ensure the results are statistically significant.
 - The more iterations, the more accurate and reliable the results.

Steps in Monte Carlo Simulation

1. **Define the Problem:**
 - Clearly state the problem and determine the quantity to be estimated.
2. **Model the System:**
 - Create a mathematical model representing the system or process, including all relevant random variables.
3. **Generate Random Inputs:**
 - Use random number generators to produce inputs that follow the specified probability distributions.
4. **Simulate the System:**
 - Run the simulation using the random inputs and record the outcomes.
5. **Analyze the Results:**
 - Compute statistics (e.g., mean, variance) on the collected outcomes to estimate the desired quantities.

Applications of Monte Carlo Methods

1. **Finance:**

- **Option Pricing:** Use to evaluate complex financial derivatives.
- **Risk Assessment:** Estimate the probability of extreme losses.
- 2. **Physics and Engineering:**
 - **Particle Transport:** Simulate the behavior of particles passing through materials.
 - **Reliability Analysis:** Estimate the failure probability of systems and components.
- 3. **Mathematics:**
 - **Integration:** Estimate the value of integrals, especially in high dimensions where traditional methods are inefficient.
- 4. **Computer Graphics:**
 - **Ray Tracing:** Simulate the interactions of light with surfaces to create realistic images.

Real Time Analogy: Estimating Pi

A classic example of a Monte Carlo method is estimating the value of π .

1. **Problem Definition:**
 - Estimate π by calculating the area of a quarter circle inscribed in a unit square.
2. **Model the System:**
 - Consider a unit square with a quarter circle of radius 1.
3. **Generate Random Inputs:**
 - Randomly generate points (x,y) where x and y are uniformly distributed between 0 and 1.
4. **Simulate the System:**
 - Check if each point lies inside the quarter circle: $x^2 + y^2 \leq 1$.
 - Count the number of points inside the circle N_{circle} and the total number of points N_{total} .
5. **Analyze the Results:**

- The ratio $\frac{N_{\text{circle}}}{N_{\text{total}}}$ approximates $\frac{\pi}{4}$.
- Therefore, $\pi \approx 4 \times \frac{N_{\text{circle}}}{N_{\text{total}}}$.

Benefits and Limitations

Benefits:

- **Versatility:** Applicable to a wide range of problems across various fields.
- **Simplicity:** Conceptually simple and easy to implement.
- **Flexibility:** Can handle complex systems with multiple random variables.

Limitations:

- **Computationally Intensive:** Requires a large number of samples for accurate results.
- **Convergence:** Convergence to the true value can be slow, especially for high-dimensional problems.

Session No. 2

Hypothesis Testing

Hypothesis testing is a fundamental concept in statistics used to determine whether there is enough evidence to reject a null hypothesis (H_0) in favor of an alternative hypothesis (H_1). Two critical aspects of hypothesis testing are Type I and Type II errors, which reflect the possible mistakes that can occur during the testing process.

Steps in Hypothesis Testing:

1. **State the Hypotheses:**
 - **Null Hypothesis (H_0):** The default assumption that there is no effect or no difference.
 - **Alternative Hypothesis (H_1):** The assumption that there is an effect or a difference.
2. **Choose a Significance Level (α):**
 - The probability of rejecting the null hypothesis when it is true, typically set at 0.05 or 5%.
3. **Collect Data:**
 - Gather data through experiments or observations relevant to the hypotheses.
4. **Calculate a Test Statistic:**
 - Use statistical methods to compute a value that helps in making the decision, such as a z-score, t-score, or p-value.
5. **Make a Decision:**
 - Compare the test statistic to a critical value or compare the p-value to the significance level α .
 - Reject H_0 if the test statistic exceeds the critical value or if the p-value is less than α .

Type I and II errors

In statistics, a **Type I error** is a false positive conclusion, while a **Type II error** is a false negative conclusion.

Making a statistical decision always involves uncertainties, so the risks of making these errors are unavoidable in hypothesis testing.

The probability of making a Type I error is the significance level, or alpha (α), while the probability of making a Type II error is beta (β). These risks can be minimized through careful planning in your study design.

Real Time Analogy 1:

Type I vs Type II error You decide to get tested for COVID-19 based on mild symptoms. There are two errors that could potentially occur:

- **Type I error (false positive):** the test result says you have coronavirus, but you actually don't.
- **Type II error (false negative):** the test result says you don't have coronavirus, but you actually do.

Real Time Analogy 2:

Type I and Type II errors A Type I error happens when you get false positive results: you conclude that the drug intervention improved symptoms when it actually didn't. These improvements could have arisen from other random factors or measurement errors.

A Type II error happens when you get false negative results: you conclude that the drug intervention didn't improve symptoms when it actually did. Your study may have missed key indicators of improvements or attributed any improvements to other factors instead.

Type I and Type II Error		
Null hypothesis is ...	True	False
Rejected	Type I error False positive Probability = α	Correct decision True positive Probability = $1 - \beta$
Not rejected	Correct decision True negative Probability = $1 - \alpha$	Type II error False negative Probability = β

Type I error

A Type I error means rejecting the null hypothesis when it's actually true. It means concluding that results are **statistically significant** when, in reality, they came about purely by chance or because of unrelated factors.

The risk of committing this error is the significance level (alpha or α) you choose. That's a value that you set at the beginning of your study to assess the statistical probability of obtaining your results (p value).

The significance level is usually set at 0.05 or 5%. This means that your results only have a 5% chance of occurring, or less, if the null hypothesis is actually true.

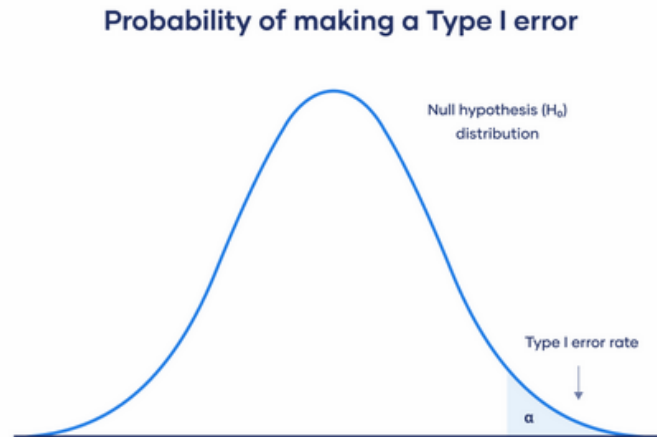
If the p value of your test is lower than the significance level, it means your results are statistically significant and consistent with the alternative hypothesis. If your p value is higher than the significance level, then your results are considered statistically non-significant.

Type I error rate

The null hypothesis distribution curve below shows the probabilities of obtaining all possible results if the study were repeated with new samples and the null hypothesis were true in the population.

At the tail end, the shaded area represents alpha. It's also called a **critical region** in statistics.

If your results fall in the critical region of this curve, they are considered statistically significant and the null hypothesis is rejected. However, this is a false positive conclusion, because the null hypothesis is actually true in this case!



Type II error

A Type II error means not rejecting the null hypothesis when it's actually false. This is not quite the same as “accepting” the null hypothesis, because hypothesis testing can only tell you whether to reject the null hypothesis.

Instead, a Type II error means failing to conclude there was an effect when there actually was. In reality, your study may not have had enough statistical power to detect an effect of a certain size.

Power is the extent to which a test can correctly detect a real effect when there is one. A power level of 80% or higher is usually considered acceptable.

The risk of a Type II error is inversely related to the statistical power of a study. The higher the statistical power, the lower the probability of making a Type II error.

Statistical power is determined by:

- **Size of the effect:** Larger effects are more easily detected.
- **Measurement error:** Systematic and random errors in recorded data reduce power.
- **Sample size:** Larger samples reduce sampling error and increase power.
- **Significance level:** Increasing the significance level increases power.

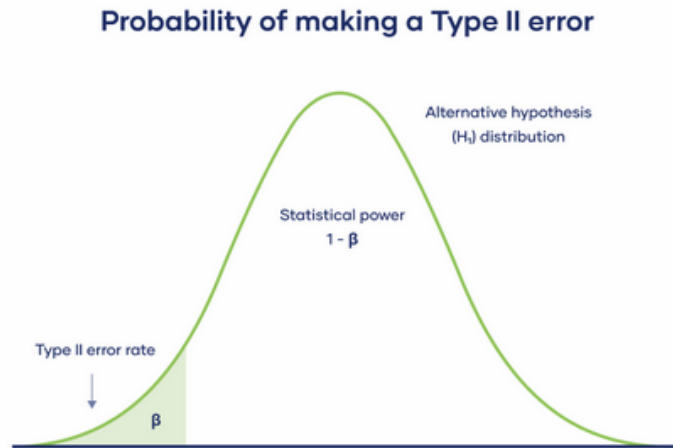
To (indirectly) reduce the risk of a Type II error, you can increase the sample size or the significance level.

Type II error rate

The alternative hypothesis distribution curve below shows the probabilities of obtaining all possible results if the study were repeated with new samples and the alternative hypothesis were true in the population.

The Type II error rate is beta (β), represented by the shaded area on the left side. The remaining area under the curve represents statistical power, which is $1 - \beta$.

Increasing the statistical power of your test directly decreases the risk of making a Type II error.



Trade-off between Type I and Type II errors

The Type I and Type II error rates influence each other. That's because the significance level (the Type I error rate) affects statistical power, which is inversely related to the Type II error rate.

This means there's an important tradeoff between Type I and Type II errors:

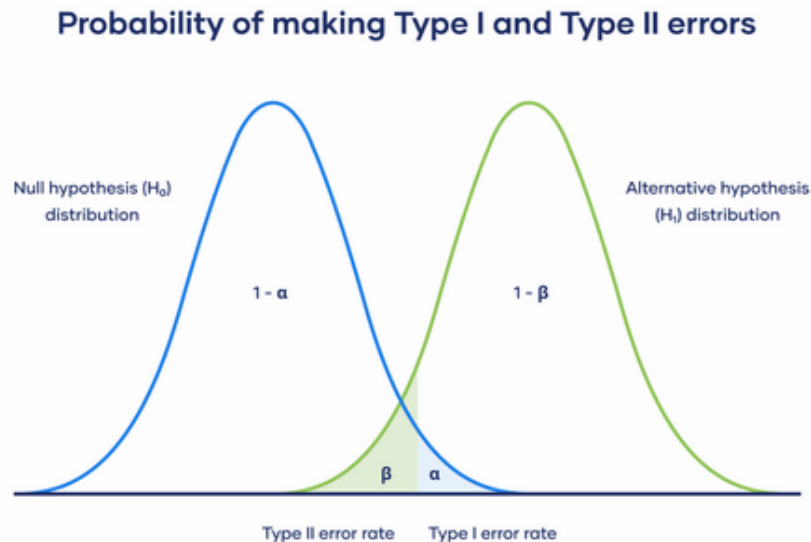
- Setting a lower significance level decreases a Type I error risk, but increases a Type II error risk.
- Increasing the power of a test decreases a Type II error risk, but increases a Type I error risk.

This trade-off is visualized in the graph below. It shows two curves:

- The **null hypothesis distribution** shows all possible results you'd obtain if the null hypothesis is true. The correct conclusion for any point on this distribution means not rejecting the null hypothesis.
- The **alternative hypothesis distribution** shows all possible results you'd obtain if the alternative hypothesis is true. The correct conclusion for any point on this distribution means rejecting the null hypothesis.

Type I and Type II errors occur where these two distributions overlap. The blue shaded area represents alpha, the Type I error rate, and the green shaded area represents beta, the Type II error rate.

By setting the Type I error rate, you indirectly influence the size of the Type II error rate as well.



It's important to strike a balance between the risks of making Type I and Type II errors. Reducing the alpha always comes at the cost of increasing beta, and vice versa.

Is a Type I or Type II error worse?

For statisticians, a Type I error is usually worse. In practical terms, however, either type of error could be worse depending on your research context.

A Type I error means mistakenly going against the main statistical assumption of a null hypothesis. This may lead to new policies, practices or treatments that are inadequate or a waste of resources.

Example: Consequences of a Type I errorBased on the incorrect conclusion that the new drug intervention is effective, over a million patients are prescribed the medication, despite risks of severe side effects and inadequate research on the outcomes. The consequences of this Type I error also mean that other treatment options are rejected in favor of this intervention.

In contrast, a Type II error means failing to reject a null hypothesis. It may only result in missed opportunities to innovate, but these can also have important practical consequences.

Example: Consequences of a Type II errorIf a Type II error is made, the drug intervention is considered ineffective when it can actually improve symptoms of the disease. This means that a medication with important clinical significance doesn't reach a large number of patients who could tangibly benefit from it.

Session No. 3

Rejection Regions

In hypothesis testing, the rejection region is a range of values for the test statistic that leads to the rejection of the null hypothesis (H_0). It is determined by the chosen significance level (α). If the test statistic falls within the rejection region, H_0 is rejected in favor of the alternative hypothesis (H_1).

Z-test

The Z-test is a statistical method used to determine whether there is a significant difference between sample data and a population parameter or between two sample means. It is typically used when the sample size is large ($n > 30$) and the population variance is known. Here's a detailed explanation of the Z-test:

Types of Z-tests

1. **One-Sample Z-test:** Tests whether the mean of a single sample is significantly different from a known population mean.
2. **Two-Sample Z-test:** Tests whether the means of two independent samples are significantly different from each other.
3. **Proportion Z-test:** Tests whether the proportion of successes in a single sample is significantly different from a known population proportion or whether the proportions in two samples are different.

Assumptions of Z-test

1. **Normal Distribution:** The data should follow a normal distribution. For large samples ($n > 30$), the Central Limit Theorem ensures that the sampling distribution of the mean will be approximately normal.
2. **Known Population Variance:** The population variance should be known. If not, the T-test is more appropriate.

One-Sample Z-test

Hypotheses

- **Null Hypothesis (H_0):** The sample mean is equal to the population mean ($\mu = \mu_0$).
- **Alternative Hypothesis (H_1):** The sample mean is not equal to the population mean ($\mu \neq \mu_0$), greater than ($\mu > \mu_0$), or less than ($\mu < \mu_0$).

Test Statistic

The test statistic for a one-sample Z-test is calculated using the formula:

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Where:

- $\bar{X}$ is the sample mean
- μ_0 is the population mean
- σ is the population standard deviation
- n is the sample size

Two-Sample Z-test

Hypotheses

- **Null Hypothesis (H0):** The means of two samples are equal ($\mu_1 = \mu_2$).
- **Alternative Hypothesis (H1):** The means of two samples are not equal ($\mu_1 \neq \mu_2$), greater ($\mu_1 > \mu_2$), or less ($\mu_1 < \mu_2$).

Test Statistic

The test statistic for a two-sample Z-test is calculated using the formula:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

Where:

- $\bar{X}_1$ and $\bar{X}_2$ are the sample means
- μ_1 and μ_2 are the population means
- σ_1 and σ_2 are the population standard deviations
- n_1 and n_2 are the sample sizes

Real Time Application of a One-Sample Z-test

Suppose a factory claims that its light bulbs have a mean life of 1000 hours. A random sample of 50 light bulbs has a mean life of 980 hours and a population standard deviation of 30 hours. We want to test if the mean life of the bulbs is different from the claimed 1000 hours at a 5% significance level.

1. **State the Hypotheses:**
 - $H_0: \mu = 1000$
 - $H_1: \mu \neq 1000$
2. **Choose Significance Level (α):**
 - $\alpha = 0.05$
3. **Compute the Test Statistic (Z):**

$$Z = \frac{980 - 1000}{30/\sqrt{50}} = \frac{-20}{4.24} \approx -4.72$$

1. Determine the Critical Value:

- For a two-tailed test with $\alpha=0.05$, the critical values are ± 1.96 .

2. Make a Decision:

- Since $-4.72 > 1.96$, we reject the null hypothesis.

Thus, there is significant evidence to suggest that the mean life of the light bulbs is different from the claimed 1000 hours.

T-Test

A t test is a statistical test that is used to compare the means of two groups. It is often used in hypothesis testing to determine whether a process or treatment actually has an effect on the population of interest, or whether two groups are different from one another.

- The null hypothesis (H_0) is that the true difference between these group means is zero.
- The alternate hypothesis (H_a) is that the true difference is different from zero.

When to use a t test

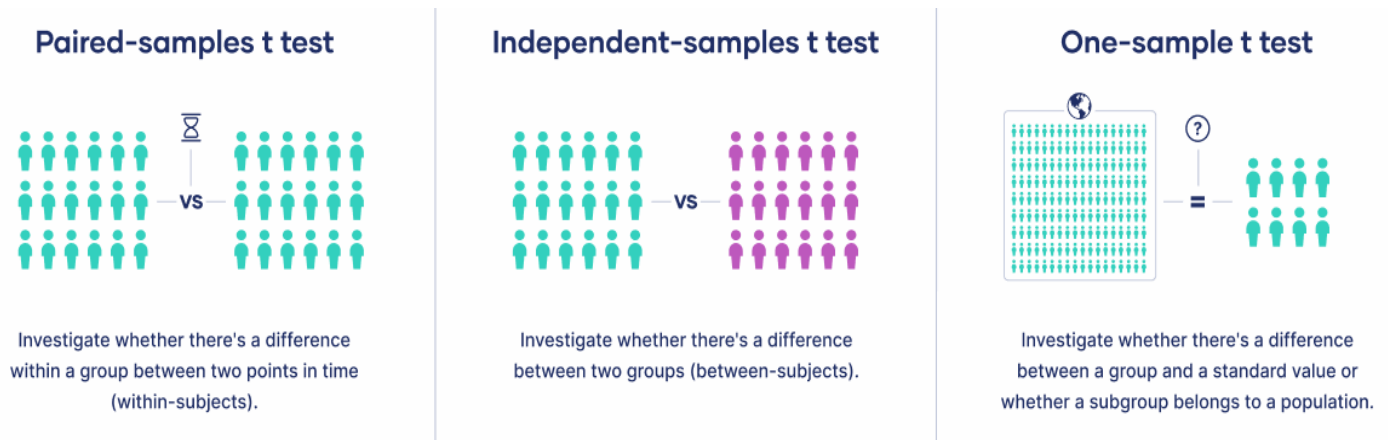
A t test can only be used when comparing the means of two groups (a.k.a. pairwise comparison). If you want to compare more than two groups, or if you want to do multiple pairwise comparisons, use an ANOVA test or a post-hoc test.

The t test is a parametric test of difference, meaning that it makes the same assumptions about your data as other parametric tests. The t test assumes your data:

1. are independent
2. are (approximately) normally distributed
3. have a similar amount of variance within each group being compared (a.k.a. homogeneity of variance)

If your data do not fit these assumptions, you can try a nonparametric alternative to the t test, such as the Wilcoxon Signed-Rank test for data with unequal variances.

When choosing a t test, you will need to consider two things: whether the groups being compared come from a single population or two different populations, and whether you want to test the difference in a specific direction.



F test

F test is a **statistical test** that is used in hypothesis testing, that determines whether or not the variances of two populations or two samples are equal. An f distribution is what the data in a f test conforms to. By dividing the two variances, this test compares them using the f statistic. Depending on the details of the situation, a f-test can be one-tailed or two-tailed. The article will provide further information on the f test, the f statistic, its calculation, critical value, and how to use it to test hypotheses.

F-distribution

In statistical hypothesis testing, the Fisher-Snedecor distribution, sometimes known as the F-distribution, is used, particularly when comparing variances or testing means across populations. Two degrees of freedom parameters, represented as df_1 (numerator) and df_2 (denominator), define this positively skewed data.

In statistical analyses such as ANOVA or regression, the F-test assesses the equality of variances or means, as it is derived from the F-distribution. In hypothesis testing, the rejection zone is determined by critical values derived from the F-distribution, taking into account the degrees of freedom and significance level. An F-distribution can be found when two independent chi-square variables are split by their corresponding degrees of freedom.

$$f\text{-value} = \text{sample1}/df_1 / \text{sample2}/df_2$$

- The independent random variables, Samples 1 and 2, have a chi-square distribution.
- The related samples' degrees of freedom are denoted by df_1 and df_2 .

Degree of Freedom

The degrees of freedom represent the number of observations used to calculate the chi-square variables that form the ratio. The shape of the F-distribution is determined by its degrees of freedom. It is a right-skewed distribution, meaning it has a longer tail on the right side. As the degrees of freedom increase, the F-distribution becomes more symmetric and approaches a bell shape.

What is F-Test?

The F test is a statistical technique that determines if the variances of two samples or populations are equal using the F test statistic. Both the samples and the populations need to be independent and fit into an F-distribution. The null hypothesis can be rejected if the results of the F test during the hypothesis test are statistically significant; if not, it stays unchanged.

We can use this test when:

- The population is normally distributed.
- The samples are taken at random and are independent samples.

F-Test Formula using two covariances

$$F_{\text{calc}} = \sigma_1^2 / \sigma_2^2$$

Here,

- F_{calc} = Critical F-value.
- σ_1^2 & σ_2^2 = variance of the two samples.

$$df = n_s - 1$$

Here,

- df = Degrees of freedom of the sample.
- n_s = Sample size.

Hypothesis Testing Framework for F-test

Using hypothesis testing, the f test is performed to verify that the variances are equivalent. For various hypothesis tests, the f test formula is provided as follows:

Left Tailed Test:

Null Hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$

Alternate Hypothesis: $H_1: \sigma_1^2 < \sigma_2^2$

Decision-Making Standard: The null hypothesis is to be rejected if the f statistic is less than the f critical value.

Right Tailed Test:

Null Hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$

Alternate Hypothesis: $H_1: \sigma_1^2 > \sigma_2^2$

Decision-Making Standard: Dismiss the null hypothesis if the f test statistic is greater than the f test critical value.

Two Tailed Test:

Null Hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$

Alternate Hypothesis: $H_1: \sigma_1^2 \neq \sigma_2^2$

Decision-Making Standard: When the f test statistic surpasses the f test critical value, the null hypothesis is declared invalid.

Real Time Analogy

Imagine you are comparing the exam scores of three different teaching methods for an AI course. You have three groups of students, each taught with a different method. You want to know if the teaching method affects the variance in exam scores.

1. **Calculate Variances:** Calculate the variance for each group.
2. **Compute the F-Statistic:** Compare the largest variance to the smallest variance.
3. **Degrees of Freedom:** For each variance, determine the degrees of freedom (number of observations minus one for each group).
4. **Compare to Critical Value:** Use an F-table to find the critical F value for your degrees of freedom at the 0.05 significance level.
5. **Make a Decision:** If your F-statistic exceeds the critical value, you conclude that the variances are significantly different, suggesting that the teaching methods affect the variability in exam scores.

The F-test is a powerful tool in statistical analysis, allowing researchers to compare variances and determine the significance of group differences or model effectiveness.

Chi-Square test

A **chi-square test** is a statistical test for categorical data. It is used to determine whether your data are significantly different from what you expected. There are two types of Pearson's chi-square tests:

- The **chi-square goodness of fit test** is used to test whether the frequency distribution of a categorical variable is different from your expectations.
- The **chi-square test of independence** is used to test whether two categorical variables are related to each other.

Chi-square is often written as X^2 and is pronounced “kai-square” (rhymes with “eye-square”). It is also called chi-squared.

What is a chi-square test?

Pearson's chi-square (X^2) tests, often referred to simply as chi-square tests, are among the most common **nonparametric tests**. Nonparametric tests are used for data that don't follow the assumptions of parametric tests, especially the assumption of a normal distribution.

If you want to test a hypothesis about the distribution of a **categorical variable** you'll need to use a chi-square test or another nonparametric test. Categorical variables can be nominal or ordinal and represent groupings such as species or nationalities. Because they can only have a few specific values, they can't have a normal distribution.

Note: Parametric tests can't test hypotheses about the distribution of a categorical variable, but they can involve a categorical variable as an independent variable (e.g., ANOVAs).

Test hypotheses about frequency distributions

There are two types of Pearson's chi-square tests, but they both test whether the observed **frequency distribution** of a categorical variable is significantly different from its expected frequency distribution. A frequency distribution describes how observations are distributed between different groups.

Frequency distributions are often displayed using **frequency distribution tables**. A frequency distribution table shows the number of observations in each group. When there are two categorical variables, you can use a specific type of frequency distribution table called a **contingency table** to show the number of observations in each combination of groups.

The chi-square formula

Both of Pearson's chi-square tests use the same formula to calculate the test statistic, chi-square (X^2):

$$X^2 = \sum \frac{(O - E)^2}{E}$$

Where:

- X^2 is the chi-square test statistic
- Σ is the summation operator (it means "take the sum of")
- O is the observed frequency
- E is the expected frequency

The larger the difference between the observations and the expectations ($O - E$ in the equation), the bigger the chi-square will be. To decide whether the difference is big enough to be statistically significant, you compare the chi-square value to a critical value.

When to use a chi-square test

A Pearson's chi-square test may be an appropriate option for your data if **all** of the following are true:

1. You want to test a hypothesis about **one or more categorical variables**. If one or more of your variables is quantitative, you should use a different statistical test. Alternatively,

you could convert the quantitative variable into a categorical variable by separating the observations into intervals.

2. The **sample was randomly selected** from the population.
3. There are a **minimum of five observations expected** in each group or combination of

Example: Handedness and nationality

	Right-handed	Left-handed
American	236	19
Canadian	157	16

A chi-square test (a test of independence) can test whether these observed frequencies are significantly different from the frequencies expected if handedness is unrelated to nationality.

Types of chi-square tests

The two types of Pearson's chi-square tests are:

- Chi-square goodness of fit test
- Chi-square test of independence

Session No. 4

Bayesian test

Bayesian inference derives the posterior probability as a consequence of two antecedents: a prior probability and a "likelihood function" derived from a statistical model for the observed data. Bayesian inference computes the posterior probability according to Bayes' theorem:

$$P(H | E) = \frac{P(E | H) \cdot P(H)}{P(E)},$$

where

- H stands for any *hypothesis* whose probability may be affected by data (called *evidence* below). Often there are competing hypotheses, and the task is to determine which is the most probable.
- $P(H)$, the *prior probability*, is the estimate of the probability of the hypothesis H *before* the data E, the current evidence, is observed.
- E, the *evidence*, corresponds to new data that were not used in computing the prior probability.
- $P(H | E)$, the *posterior probability*, is the probability of H *given* E, i.e., *after* E is observed. This is what we want to know: the probability of a hypothesis *given* the observed evidence.
- $P(E | H)$ is the **probability of observing E given H** and is called the *likelihood*. As a function of E with H fixed, it indicates the compatibility of the evidence with the given hypothesis. The likelihood function is a function of the evidence, E, while

the posterior probability is a function of the hypothesis, H .

- $P(E)$ is sometimes termed the marginal likelihood or "model evidence". This factor is the same for all possible hypotheses being considered (as is evident from the fact that the hypothesis H does not appear anywhere in the symbol, unlike for all the other factors) and hence does not factor into determining the relative probabilities of different hypotheses.
- $P(E) > 0$ (Else one has $0/0$.)

For different values of H , only the factors $P(H)$ and $P(E|H)$, both in the numerator, affect the value of $P(H|E)$ – the posterior probability of a hypothesis is proportional to its prior probability (its inherent likeliness) and the newly acquired likelihood (its compatibility with the new observed evidence).

In cases where $\neg H$ ("not H "), the logical negation of H , is a valid likelihood, Bayes' rule can be rewritten as follows:

$$\begin{aligned} P(H|E) &= \frac{P(E|H)P(H)}{P(E)} \\ &= \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|\neg H)P(\neg H)} \\ &= \frac{1}{1 + \left(\frac{1}{P(H)} - 1\right) \frac{P(E|\neg H)}{P(E|H)}} \end{aligned}$$

because

$$P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H)$$

and

$$P(H) + P(\neg H) = 1.$$

This focuses attention on the term

$$\left(\frac{1}{P(H)} - 1\right) \frac{P(E|\neg H)}{P(E|H)}.$$

If that term is approximately 1, then the probability of the hypothesis given the evidence, $P(H|E)$, is about 1/2, about 50% likely – equally likely or not likely. If that term is very small, close to zero, then the probability of the hypothesis, given the evidence, $P(H|E)$ is close to 1 or the conditional hypothesis is quite likely. If that term is very large, much larger than 1, then the hypothesis, given the evidence, is quite unlikely. If the hypothesis (without consideration of evidence) is unlikely, then $P(H)$ is small (but not necessarily astronomically small) and $1/P(H)$ is much larger than 1 and this term can be approximated as $P(E|\neg H)/P(E|H) \cdot P(H)$ and relevant probabilities can be compared directly to each other.

Stochastic Processes and Data Modeling

A stochastic process is a collection of random variables indexed by time or space, representing systems or phenomena that evolve over time in a probabilistic manner. Stochastic processes are used to model uncertainty and randomness in dynamic systems.

Key Concepts:

1. **Random Variable:** A variable whose values are outcomes of a random phenomenon.
2. **Index Set:** Typically time or space, denoted as T . For example, T could be the set of all non-negative integers for discrete-time processes or the set of all non-negative real numbers for continuous-time processes.
3. **State Space:** The set of all possible values that the random variables can take.

Types of Stochastic Processes:

1. **Discrete-Time Processes:** The index set T is discrete (e.g., $T = \{0, 1, 2, \dots\}$).
 - **Example:** A random walk, where the position changes at fixed time intervals.
2. **Continuous-Time Processes:** The index set T is continuous (e.g., $T = [0, \infty)$).
 - **Example:** Brownian motion, describing the random movement of particles suspended in a fluid.
3. **Markov Processes:** A process where the future state depends only on the present state and not on the past states (memoryless property).
 - **Example:** A simple weather model where tomorrow's weather depends only on today's weather.
4. **Poisson Processes:** A process counting the number of events in fixed intervals of time or space, where the events occur independently.
 - **Example:** The number of phone calls received at a call center per hour.

Data Modeling

Data modeling involves creating a mathematical representation of a real-world process using data. The goal is to understand the underlying patterns, make predictions, and inform decision-making. In the context of stochastic processes, data modeling helps in fitting a model to the observed data and validating its assumptions.

Key Steps in Data Modeling:

1. **Data Collection:** Gather data relevant to the phenomenon being studied.
2. **Exploratory Data Analysis (EDA):** Analyze data to understand its main characteristics, often using statistical graphics and other data visualization tools.
3. **Model Selection:** Choose a suitable model that can describe the data. This could be a stochastic process model like a Markov chain or a Poisson process.
4. **Parameter Estimation:** Estimate the parameters of the chosen model using methods like Maximum Likelihood Estimation (MLE) or Bayesian inference.
5. **Model Validation:** Check the model's adequacy using techniques such as goodness-of-fit tests, residual analysis, and cross-validation.

6. **Prediction and Inference:** Use the model to make predictions and draw inferences about the real-world process.

Real Time Application

1. **Finance:**
 - **Stochastic Process:** Modeling stock prices using geometric Brownian motion.
 - **Data Modeling:** Fitting the model to historical stock price data to forecast future prices and assess risks.

Markov process

A Markov process is a type of stochastic process that has a specific property: the future state of the process depends only on the present state and not on the sequence of events that preceded it. This property is known as the "Markov property" or "memoryless property."

Key Concepts of Markov Process

1. **State Space:** The set of all possible states that the process can be in.
2. **Transition Probability:** The probability of moving from one state to another.
3. **Markov Property:** The conditional probability of the next state depends only on the current state, not on the previous states. Mathematically, if X_t is the state at time t , then:

$$P(X_{t+1} = x_{t+1} \mid X_t = x_t, X_{t-1} = x_{t-1}, \dots, X_0 = x_0) = P(X_{t+1} = x_{t+1} \mid X_t = x_t)$$

Types of Markov Processes

1. **Discrete-Time Markov Chain (DTMC):** The process evolves in discrete time steps.
 - **Transition Matrix:** Describes the probabilities of moving from one state to another in one time step. If there are n states, the transition matrix P is an $n \times n$ matrix where P_{ij} is the probability of transitioning from state i to state j .

$$P = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1n} \\ P_{21} & P_{22} & \cdots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \cdots & P_{nn} \end{bmatrix}$$

Continuous-Time Markov Process (CTMP): The process evolves continuously over time.

- **Transition Rate Matrix (or Generator Matrix):** Describes the rate at which transitions occur between states. If there are n states, the generator matrix Q is an $n \times n$ matrix where Q_{ij} (for $i \neq j$) is the rate of moving from state i to state j , and Q_{ii} is the negative of the rate of leaving state i .

Real Time Application

Weather Modeling: A Markov chain can model weather conditions, where each state represents a type of weather (sunny, rainy, cloudy), and the transition probabilities represent the likelihood of changing from one weather state to another.

Hidden Markov Models

Hidden Markov Models (HMMs) are a type of probabilistic model that are commonly used in machine learning for tasks such as speech recognition, natural language processing, and bioinformatics. They are a popular choice for modeling sequences of data because they can effectively capture the underlying structure of the data, even when the data is noisy or incomplete. In this article, we will give a comprehensive overview of Hidden Markov Models, including their mathematical foundations, applications, and limitations.

What are Hidden Markov Models?

A Hidden Markov Model (HMM) is a probabilistic model that consists of a sequence of hidden states, each of which generates an observation. The hidden states are usually not directly observable, and the goal of HMM is to estimate the sequence of hidden states based on a sequence of observations. An HMM is defined by the following components:

- A set of N hidden states, $S = \{s_1, s_2, \dots, s_N\}$.
- A set of M observations, $O = \{o_1, o_2, \dots, o_M\}$.
- An initial state probability distribution, $\pi = \{\pi_1, \pi_2, \dots, \pi_N\}$, which specifies the probability of starting in each hidden state.
- A transition probability matrix, $A = [a_{ij}]$, defines the probability of moving from one hidden state to another.
- An emission probability matrix, $B = [b_{jk}]$, defines the probability of emitting an observation from a given hidden state.

The basic idea behind an HMM is that the hidden states generate the observations, and the observed data is used to estimate the hidden state sequence. This is often referred to as the **forward-backwards algorithm**.

Real Time Analogy of Hidden Markov Models

- **Speech Recognition**

One of the most well-known applications of HMMs is speech recognition. In this field, HMMs are used to model the different sounds and phones that makeup speech. The hidden states, in this case, correspond to the different sounds or phones, and the observations are the acoustic signals that are generated by the speech. The goal is to estimate the hidden state sequence, which corresponds to the transcription of the speech, based on the observed acoustic signals. HMMs are particularly well-suited for

speech recognition because they can effectively capture the underlying structure of the speech, even when the data is noisy or incomplete. In speech recognition systems, the HMMs are usually trained on large datasets of speech signals, and the estimated parameters of the HMMs are used to transcribe speech in real time.

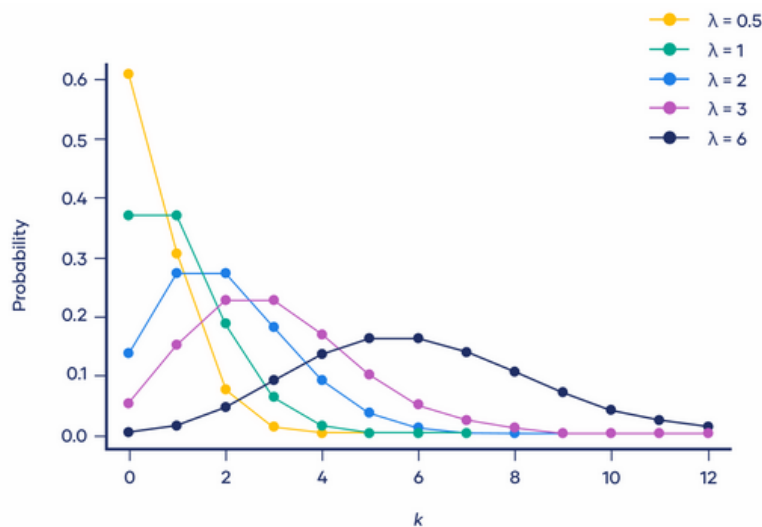
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Poisson Process

A Poisson process is a stochastic process that models a series of events occurring randomly over time. It is characterized by its simplicity and wide applicability in various fields such as queuing theory, telecommunications, finance, and natural sciences.

A **Poisson distribution** is a discrete probability distribution. It gives the probability of an event happening a certain number of times (k) within a given interval of time or space.

The Poisson distribution has only one parameter, λ (lambda), which is the mean number of events. The graph below shows examples of Poisson distributions with different values of λ .



What is a Poisson distribution?

A Poisson distribution is a discrete probability distribution, meaning that it gives the probability of a discrete (i.e., countable) outcome. For Poisson distributions, the discrete outcome is the number of times an event occurs, represented by k .

You can use a Poisson distribution to predict or explain the number of events occurring within a given interval of time or space. “Events” could be anything from disease cases to customer purchases to meteor strikes. The interval can be any specific amount of time or space, such as 10 days or 5 square inches.

Poisson distribution formula

The probability mass function of the Poisson distribution is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Where:

- X is a random variable following a Poisson distribution
- k is the number of times an event occurs
- $P(X = k)$ is the probability that an event will occur k times
- e is Euler's constant (approximately 2.718)
- λ is the average number of times an event occurs
- $!$ is the factorial function

Real Time Application

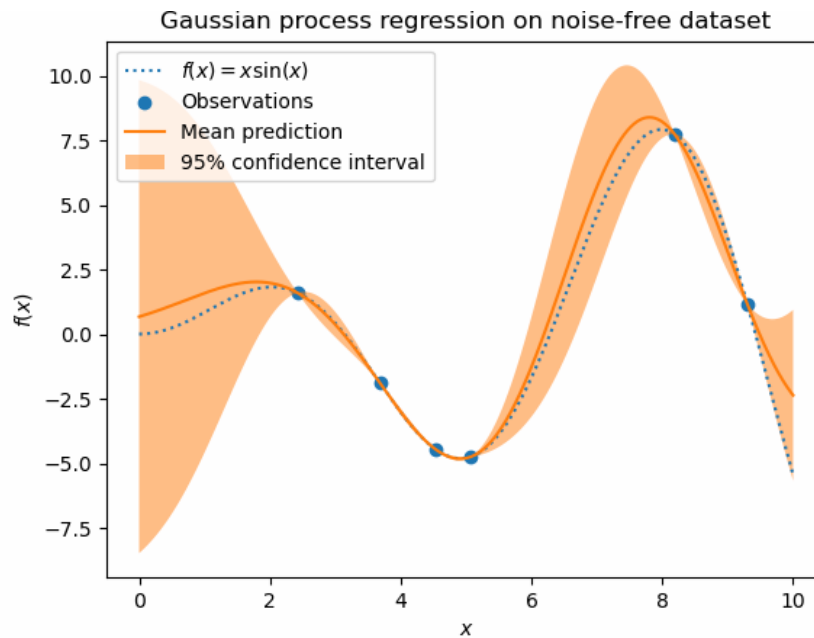
Example: Applying the Poisson distribution formula An average of 0.61 soldiers died by horse kicks per year in each Prussian army corps. You want to calculate the probability that exactly two soldiers died in the VII Army Corps in 1898, assuming that the number of horse kick deaths per year follows a Poisson distribution.

Gaussian Processes

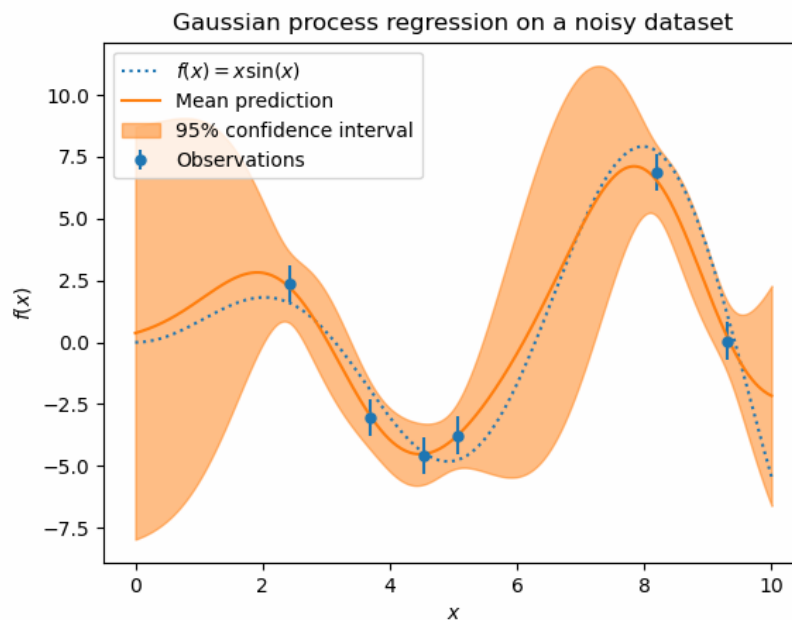
Gaussian Processes (GP) are a nonparametric supervised learning method used to solve *regression* and *probabilistic classification* problems.

Gaussian Process Regression (GPR)

The GaussianProcessRegressor implements Gaussian processes (GP) for regression purposes. For this, the prior of the GP needs to be specified. GP will combine this prior and the likelihood function based on training samples. It allows a probabilistic approach to prediction by giving the mean and standard deviation as output when predicting.



The noise level in the targets can be specified by passing it via the parameter `alpha`, either globally as a scalar or per datapoint. Note that a moderate noise level can also be helpful for dealing with numeric instabilities during fitting as it is effectively implemented as Tikhonov regularization, i.e., by adding it to the diagonal of the kernel matrix. An alternative to specifying the noise level explicitly is to include a `WhiteKernel` component into the kernel, which can estimate the global noise level from the data (see example below). The figure below shows the effect of noisy targets handled by setting the parameter `alpha`.



Gaussian Process Classification (GPC)

The `GaussianProcessClassifier` implements Gaussian processes (GP) for classification purposes, more specifically for probabilistic classification, where test predictions take the form of class probabilities. The Gaussian Process Classifier places a GP prior on a latent function, which is

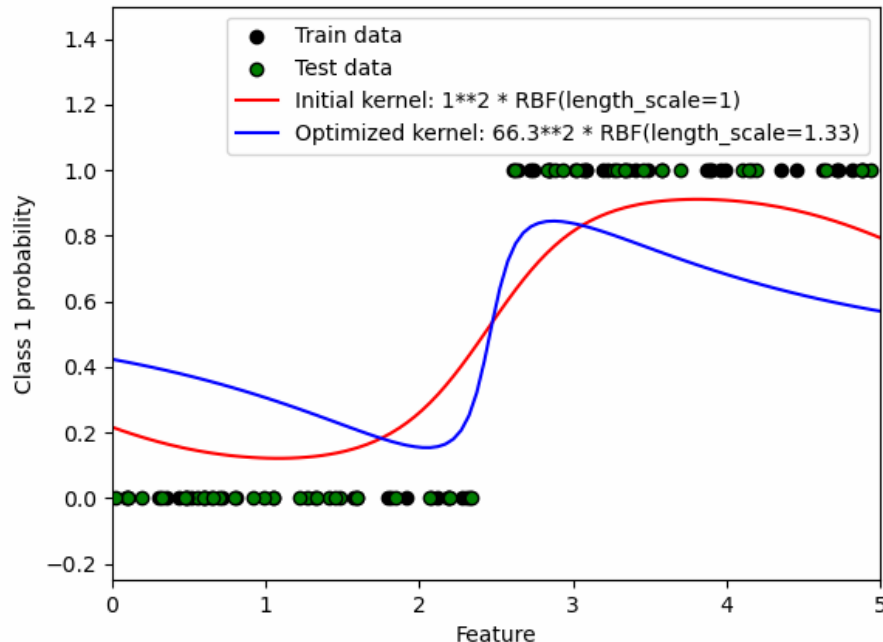
then squashed through a link function to obtain the probabilistic classification. The latent function is a so-called nuisance function, whose values are not observed and are not relevant by themselves. Its purpose is to allow a convenient formulation of the model, and is removed (integrated out) during prediction. GaussianProcessClassifier implements the logistic link function, for which the integral cannot be computed analytically but is easily approximated in the binary case.

Real Time Analogy of GPC

Probabilistic predictions with GPC

This example illustrates the predicted probability of GPC for an RBF kernel with different choices of the hyperparameters. The first figure shows the predicted probability of GPC with arbitrarily chosen hyperparameters and with the hyperparameters corresponding to the maximum log-marginal-likelihood (LML).

While the hyperparameters chosen by optimizing LML have a considerably larger LML, they perform slightly worse according to the log-loss on test data. The figure shows that this is because they exhibit a steep change of the class probabilities at the class boundaries (which is good) but have predicted probabilities close to 0.5 far away from the class boundaries (which is bad). This undesirable effect is caused by the Laplace approximation used internally by GPC.



Auto-Regressive and Moving average processes

Auto-Regressive (AR) and Moving Average (MA) processes are fundamental concepts in time series analysis, used to model and predict time-dependent data. Both are specific types of linear stochastic processes.

Auto-Regressive (AR) Processes

An Auto-Regressive (AR) process models the current value of a time series as a linear combination of its previous values plus a stochastic error term.

AR(p) Model

The notation AR(p) refers to an auto-regressive model of order p, where p indicates the number of lagged values used. The model is defined as:

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t$$

where:

- X_t is the value of the time series at time t.
- c is a constant term.
- ϕ_i are the parameters of the model.
- ϵ_t is white noise error term (a sequence of uncorrelated random variables with mean zero and constant variance).

Example: AR(1) Model

An AR(1) model (first-order auto-regressive) is given by:

$$X_t = c + \phi_1 X_{t-1} + \epsilon_t$$

This means the current value X_t depends on the previous value X_{t-1} and a stochastic error ϵ_t .

Properties of AR Processes

1. **Stationarity:** For an AR process to be stationary, the roots of the characteristic equation must lie outside the unit circle.
2. **Autocorrelation Function (ACF):** The ACF of an AR(p) process tails off geometrically.
3. **Partial Autocorrelation Function (PACF):** The PACF of an AR(p) process will have non-zero values up to lag p and zero thereafter.

Moving Average (MA) Processes

A Moving Average (MA) process models the current value of a time series as a linear combination of past white noise error terms.

MA(q) Model

The notation MA(q) refers to a moving average model of order q, where q indicates the number of lagged error terms used. The model is defined as:

$$X_t = \mu + \epsilon_t + \sum_{i=1}^q \theta_i \epsilon_{t-i}$$

This means the current value X_t depends on the current white noise ϵ_t and the previous white noise ϵ_{t-1} .

Properties of MA Processes

1. **Stationarity:** MA processes are always stationary because they are finite linear combinations of white noise terms.
2. **Autocorrelation Function (ACF):** The ACF of an MA(q) process will have non-zero values up to lag q and zero thereafter.
3. **Partial Autocorrelation Function (PACF):** The PACF of an MA(q) process tails off geometrically.

ARMA Processes

Often, a combination of AR and MA processes, called ARMA (Auto-Regressive Moving Average) models, is used for more complex time series data. An ARMA(p, q) model is defined as:

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t + \sum_{j=1}^q \theta_j \epsilon_{t-j}$$

where:

- p is the order of the AR part.
- q is the order of the MA part.
- The other terms are as previously defined.

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Bayesian Network

A Bayesian Network, also known as a Belief Network or a Directed Acyclic Graphical Model, is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG). Bayesian Networks are used for modeling the probabilistic relationships among a set of variables, facilitating reasoning under uncertainty.

"A Bayesian network is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph."

It is also called a **Bayes network**, **belief network**, **decision network**, or **Bayesian model**.

Bayesian networks are probabilistic, because these networks are built from a **probability distribution**, and also use probability theory for prediction and anomaly detection.

Real world applications are probabilistic in nature, and to represent the relationship between multiple events, we need a Bayesian network. It can also be used in various tasks including

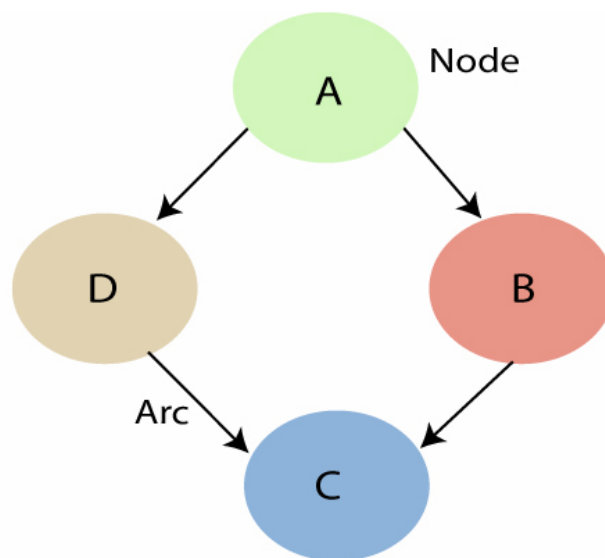
prediction, anomaly detection, diagnostics, automated insight, reasoning, time series prediction, and decision making under uncertainty.

Bayesian Network can be used for building models from data and experts opinions, and it consists of two parts:

- **Directed Acyclic Graph**
- **Table of conditional probabilities.**

The generalized form of Bayesian network that represents and solve decision problems under uncertain knowledge is known as an **Influence diagram**.

A Bayesian network graph is made up of nodes and Arcs (directed links), where:



- Each node corresponds to the random variables, and a variable can be continuous or discrete.
- Arc or directed arrows represent the causal relationship or conditional probabilities between random variables. These directed links or arrows connect the pair of nodes in the graph.

These links represent that one node directly influence the other node, and if there is no directed link that means that nodes are independent with each other

- In the above diagram, A, B, C, and D are random variables represented by the nodes of the network graph.
- If we are considering node B, which is connected with node A by a directed arrow, then node A is called the parent of Node B.
- Node C is independent of node A.

Real Time Application

Example: Harry installed a new burglar alarm at his home to detect burglary. The alarm reliably responds at detecting a burglary but also responds for minor earthquakes. Harry has two neighbors David and Sophia, who have taken a responsibility to inform Harry at work when they hear the alarm. David always calls Harry when he hears the alarm, but sometimes he gets confused with the phone ringing and calls at that time too. On the other hand, Sophia likes to listen to high music, so sometimes she misses hearing the alarm. Here we would like to compute the probability of Burglary Alarm.

Regression

What is Regression?

Regression is a statistical approach used to analyze the relationship between a dependent variable (target variable) and one or more independent variables (predictor variables). The objective is to determine the most suitable function that characterizes the connection between these variables.

It seeks to find the best-fitting model, which can be utilized to make predictions or draw conclusions.

Regression Types

There are two main types of regression:

- **Simple Regression**
 - Used to predict a continuous dependent variable based on a single independent variable.
 - Simple linear regression should be used when there is only a single independent variable.
- **Multiple Regression**
 - Used to predict a continuous dependent variable based on multiple independent variables.
 - Multiple linear regression should be used when there are multiple independent variables.
- **NonLinear Regression**
 - Relationship between the dependent variable and independent variable(s) follows a nonlinear pattern.
 - Provides flexibility in modeling a wide range of functional forms.

Regression Algorithms

There are many different types of regression algorithms, but some of the most common include:

- **Linear Regression**
 - **Linear regression** is one of the simplest and most widely used statistical models. This assumes that there is a linear relationship between the independent and dependent variables. This means that the change in the dependent variable is proportional to the change in the independent variables.
- **Polynomial Regression**
 - **Polynomial regression** is used to model nonlinear relationships between the dependent variable and the independent variables. It adds polynomial terms to the linear regression model to capture more complex relationships.
- **Support Vector Regression (SVR)**
 - **Support vector regression (SVR)** is a type of regression algorithm that is based on the support vector machine (SVM) algorithm. SVM is a type of algorithm that is used for classification tasks, but it can also be used for regression tasks. SVR works by finding a hyperplane that minimizes the sum of the squared residuals between the predicted and actual values.
- **Decision Tree Regression**
 - **Decision tree regression** is a type of regression algorithm that builds a decision tree to predict the target value. A decision tree is a tree-like structure that consists of nodes and branches. Each node represents a decision, and each branch represents the outcome of that decision. The goal of decision tree regression is to build a tree that can accurately predict the target value for new data points.
- **Random Forest Regression**
 - **Random forest regression** is an ensemble method that combines multiple decision trees to predict the target value. Ensemble methods are a type of machine learning algorithm that combines multiple models to improve the performance of the overall model. Random forest regression works by building a large number of decision trees, each of which is trained on a different subset of the training data. The final prediction is made by averaging the predictions of all of the trees.

Regularized Linear Regression Techniques

- **Ridge Regression**
 - Ridge regression is a type of linear regression that is used to prevent overfitting. Overfitting occurs when the model learns the training data too well and is unable to generalize to new data.
- **Lasso regression**
 - Lasso regression is another type of linear regression that is used to prevent overfitting. It does this by adding a penalty term to the loss function that forces the model to use some weights and to set others to zero.

Regression Evaluation Metrics

Here are some most popular evaluation metrics for regression:

- **Mean Absolute Error (MAE):** The average absolute difference between the predicted and actual values of the target variable.

- **Mean Squared Error (MSE):** The average squared difference between the predicted and actual values of the target variable.
- **Root Mean Squared Error (RMSE):** The square root of the mean squared error.
- **Huber Loss:** A hybrid loss function that transitions from MAE to MSE for larger errors, providing balance between robustness and MSE's sensitivity to outliers.
- Root Mean Squared Logarithmic Error
- R2 – Score: Higher values indicate better fit, ranging from 0 to 1.

Real Time Applications of Regression

- **Predicting prices:** For example, a regression model could be used to predict the price of a house based on its size, location, and other features.

Queuing systems

Queuing systems, also known as queueing theory, study the behavior of waiting lines or queues. This field of applied probability is used to model the process of customers arriving, waiting in line, and being served by servers. Queuing theory is crucial in many areas such as telecommunications, traffic engineering, computing, and manufacturing.

Basic Components of Queuing Systems

1. **Customers:** Entities that arrive to be served. They can be people, data packets, machines, etc.
2. **Servers:** Entities that provide service to customers. There can be one or multiple servers.
3. **Queue:** The waiting line where customers wait if all servers are busy.
4. **Arrival Process:** The pattern in which customers arrive at the queue, often modeled by a Poisson process.
5. **Service Process:** The time it takes for a server to serve a customer, often modeled by an exponential distribution.
6. **Queue Discipline:** The order in which customers are served, such as First-In-First-Out (FIFO), Last-In-First-Out (LIFO), or priority-based.

Types of Queuing Systems

Queuing systems are often denoted by the Kendall notation $A/B/cA/B/cA/B/c$, where:

- A describes the arrival process.
- B describes the service time distribution.
- c describes the number of servers.

Common models include:

1. **M/M/1 Queue:**
 - **M:** Poisson arrival process (Markovian).
 - **M:** Exponential service time distribution (Markovian).
 - **1:** Single server.
2. This is the simplest and most commonly analyzed queuing system. It assumes that customers arrive at a Poisson rate λ and service times are exponentially distributed with rate μ .
3. **M/M/c Queue:**
 - Similar to M/M/1 but with c servers.
4. **M/G/1 Queue:**
 - **M:** Poisson arrival process.
 - **G:** General service time distribution.
 - **1:** Single server.
5. **G/G/1 Queue:**
 - **G:** General arrival process.
 - **G:** General service time distribution.
 - **1:** Single server.

Applications of Queuing Systems

1. **Telecommunications:** Managing data packets in network routers and switches.
2. **Manufacturing:** Optimizing production lines and minimizing downtime.
3. **Healthcare:** Reducing patient wait times in hospitals and clinics.
4. **Customer Service:** Improving service efficiency in call centers and retail stores.
5. **Transportation:** Managing traffic flow and reducing congestion.